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Paleomagnetic results from Tertiary volcanic strata and intrusions, Absaroka Volcanic Supergroup, Yellowstone National Park and vicinity: Contributions to the North American apparent polar wander path

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ABSTRACT

We report paleomagnetic and rock magnetic data from volcanic, volcaniclastic, and intrusive rocks of the 55–44 Ma Absaroka Volcanic Supergroup (AVS) exposed along the northeastern margin of Yellowstone National Park and adjacent areas. Demagnetization behavior and rock magnetic experiments indicate that the remanence in most samples is carried by low-Ti titanomagnetite, although high-coercivity phases are present in oxidized basalt flows. Paleomagnetic demagnetization and rock magnetic characteristics, the presence of normal and reverse polarity sites, consistency with previous results, and positive conglomerate tests suggest that the observed remanences are primary thermoremanent magnetizations of Eocene age (c. 50 Ma). An in situ grand-mean for 22 individual site- or cooling-unit means from this study that yield acceptable data combined with published data from Independence volcano yields a declination of 347.6° and inclination of 59.2° ($k = 21.8$, $\alpha_{95} = 6.8°$) and a positive reversal test. Averaging 21 virtual geomagnetic poles (VGPs) that are well-grouped yields a mean at 137.1°E, 82.5°N ($K = 17.6$, $A_{95} = 7.8°$), similar to results previously obtained from published studies from the AVS. Combining the VGPs from our study with published data yields a combined AVS pole at 146.3°E, 83.1°N ($K = 13.5$, $A_{95} = 6.2°$, $N = 42$ VGPs). Both poles are indistinguishable from c. 50 Ma cratonic and synthetic reference poles for North America, and demonstrate the relative stability of this part of the Cordillera with respect to the craton.

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1. Introduction

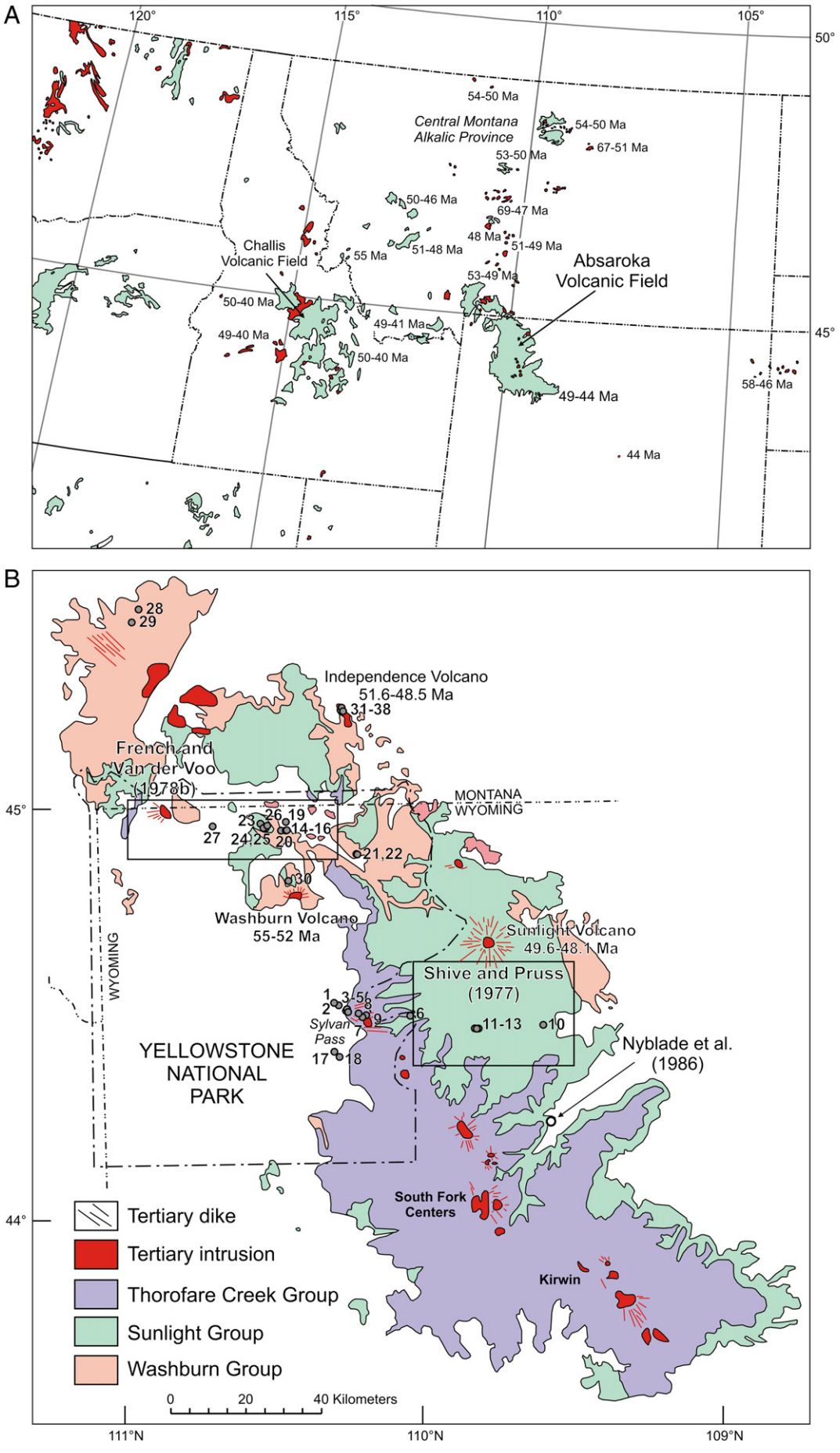
During 2000, new high-resolution aeromagnetic data from Yellowstone National Park (YNP) were processed in order to provide geophysical data useful in facilitating mapping of geologic units, structures, and areas of hydrothermal alteration within the park. To complement that investigation we collected paleomagnetic samples from Tertiary and Quaternary volcanic and intrusive rocks in YNP and surrounding areas in order to better understand sources contributing to magnetic variations observed in aeromagnetic maps published by Finn and Morgan (2002), as well as ongoing studies of the geology of Yellowstone National Park that utilize aeromagnetic data (e.g., Morgan et al., 2003a,b; Morgan et al., 2007). In this paper, we report new paleomagnetic and rock magnetic data from 30 sites in volcanic, intrusive, and sedimentary rocks of the 55–44 Ma Absaroka Volcanic Supergroup (AVS), exposed along the northern and eastern margins of YNP and surrounding areas. These results are combined with published paleomagnetic results to provide data that are valuable in understanding and interpreting magnetic anomalies observed in the aeromagnetic data for YNP and surrounding areas, provide polarity determinations that are useful in assessing stratigraphic correlations of geologic map units within the AVS, and contributes well-defined, paleomagnetic directional and pole data for the North American apparent polar wander path.

2. Geologic setting of the Absaroka volcanic field

The Absaroka volcanic field of northwest Wyoming and southwest Montana is the largest Tertiary volcanic field of the northern Rocky Mountains (Fig. 1A). It is part of a diffuse, but locally intense magmatic episode that resulted from Late Cretaceous/early Tertiary subduction of the Farallon and/or Kula plates beneath the North American plate (Atwater, 1970; Engebretson et al., 1985; Lonsdale, 1988; Madsen et al., 2006; Duke, 2009), and which affected a broad area of the northern Cordillera from ~54 to 37 Ma. This episode has been termed the Kamloops–Challis–Absaroka magmatic episode (Armstrong and Ward, 1991); temporally, it coincides with the peak of alkaline magmatism in the Central Montana Alkalic Province. The Absaroka volcanic field consists of a series of deeply eroded, coalescing stratovolcanoes of basaltic to andesitic composition, shield volcanoes, and associated

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A
120°
115°
110°
105°
50°
45°
54-50 Ma
Central Montana
Alkalic Province
54-50 Ma
67-51 Ma
53-50 Ma
50-46 Ma
69-47 Ma
48 Ma
51-49 Ma
51-48 Ma
55 Ma
Challis
Volcanic Field
53-49 Ma
Absaroka
Volcanic Field
50-40 Ma
49-41 Ma
49-40 Ma
50-40 Ma
49-44 Ma
58-46 Ma
44 Ma
B
28
29
Independence Volcano
51.6-48.5 Ma
31-38
French and
Van der Voo
(1978b)
MONTANA
WYOMING
45°
23
26
19
14-16
27
24,25
20
21,22
30
Washburn Volcano
55-52 Ma
Sunlight Volcano
49.6-48.1 Ma
WYOMING
Shive and Pruss
(1977)
1
3
5
2
8
9
6
11-13
10
Sylvan
Pass
7
YELLOWSTONE
NATIONAL
PARK
17
18
Nyblade et al.
(1986)
South Fork
Centers
44°
Kirwin
Tertiary dike
Tertiary intrusion
Thorofare Creek Group
Sunlight Group
Washburn Group
0
20
40 Kilometers
111°N
110°N
109°N

aprons of volcaniclastic and epiclastic debris (Fig. 1B). It covers an area of about 23,000 km^2, a maximum thickness in any one place of about 1.5 km, and an estimated volume of 29,000 km^3 (Smedes and Protska, 1972). Area and volume estimates are minimum values as much of the field is deeply eroded and covered by Quaternary rhyolitic rocks of the Yellowstone Plateau. Volcanic rocks consist of calc-alkaline andesite to dacite lavas with some potassic andesites, intercalated basalt flows, and minor rhyolite ignimbrite flows and sheets. Pyroclastic and epiclastic deposits consist of chaotic assemblages of proximal vent facies autoclastic flow breccias, volcanic debris flows, and tuffs that grade into or interfinger with distal volcanic conglomerates, volcanic sandstones and siltstones (Smedes and Protska, 1972; Sundell, 1993). The volcanic and volcaniclastic rocks are subdivided into three main stratigraphic groups, which in ascending order are termed the Washburn Group, the Sunlight Group, and the Thorofare Creek Group (Smedes and Protska, 1972), which together comprise the Eocene AVS (Fig. 1B). Stocks, plugs, laccoliths, and radial dike swarms are common throughout the field; together with the vent facies strata, they are thought to represent the eroded remnants of 10 to 13 individual volcanic centers (e.g. Sunlight volcano, Mount Washburn volcano, etc.). Detailed reports summarizing the geochemistry of AVS rocks include Meen and Eggler (1987); Hiza, 1999; Feeley et al. (2002); Feeley (2003); Feeley and Cosca (2003); and Lindsay and Feeley (2003).

AVS volcanism occurred along two subparallel, northwest-trending belts defined by hypabyssal plutons that extend greater than 240 km (Chadwick, 1970). The belts are inferred to have resulted from magma intrusion along Precambrian crustal weaknesses, perhaps related to northwest-striking fault zones that formed during Mesoproterozoic extension accompanying development of the Belt Basin (Chadwick, 1970; Harrison et al., 1974; Schmidt and Garihan, 1986; Sundell, 1993; Harlan et al., 1996a; and Sims et al., 2004). Geologic mapping, stratigraphic relations, and isotopic dating indicate that magmatism began as early as 55 to 53 Ma in the northwestern part of the field and migrated southeast where volcanic and intrusive rocks in the southern part of the field yield dates of ~46 to 44 Ma (Chadwick, 1970; Smedes and Protska, 1972; Harlan et al., 1996b; Hiza, 1999; Feeley et al., 2002; Feeley and Cosca, 2003; Kellogg and Harlan, 2007), although peak activity was between 50.5 and 47.5 Ma (Fig. 2). AVS rocks largely postdate Laramide (Late Cretaceous-early Tertiary) deformation and in many places rest nonconformably on Archean crystalline basement rocks or moderate to steeply-tilted Paleozoic to lower Cenozoic sedimentary strata (Smedes and Protska, 1972; Roberts, 1972). Although the exact relationship between magmatism and plate tectonic processes remains enigmatic, the eruption of the AVS rocks and other middle Tertiary volcanic and intrusive rocks in the northern Cordillera is consonant with the onset of crustal extension that post-dated Sevier and Laramide contraction. With the exception of AVS rocks in the upper plate of the Heart Mountain detachment (Pierce, 1987; Hauge, 1993), volcanic rocks with the Absaroka volcanic field have suffered little, if any, post-emplacement deformation. Although dips in vent facies rocks may locally be as high as 20° to 30°, they are considered to be primary, and some intrusive centers (e.g., Independence volcano) have quaquaversal strata (Rubel, 1971). Distal facies sedimentary rocks typically have dips less than 5°.

3. Previous paleomagnetic studies of the Absaroka Volcanic Supergroup

Despite the large size of the volcanic field, there have only been a few paleomagnetic studies of AVS rocks, mostly in reconnaissance

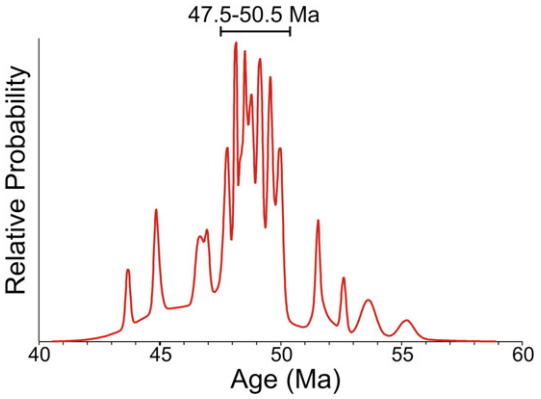


Fig. 2. Relative probability diagram showing the distribution of 63 K–Ar and $^{40}Ar/^{39}Ar$ dates from 47 individual flows or intrusions within the Absaroka volcanic field. Data compiled using data from Smedes and Protska (1972); Chadwick (1970, 1980); Bown (1982); Luedke (1994); Harlan et al. (1996b); Hiza (1999); Feeley et al. (2002); Feeley and Cosca (2003); Kellogg and Harlan (2007) and Taylor et al. (2007).

fashion. This is due to the extremely steep and rugged terrain and difficult access to remote areas. In addition, much of the area in which the volcanics are exposed is within protected wilderness areas or Yellowstone National Park (YNP). Detailed paleomagnetic results that include directional data were reported by Shive and Pruss (1977), Nyblade et al. (1986), and Harlan et al. (1996b). In addition, preliminary results were published in abstract and summary form by French and Van der Voo (1978a,b) and Moose and Acton (1993). Summary paleomagnetic results (mostly polarity data) in support of stratigraphic or volcanological studies of the AVS rocks are described by Smedes and Protska (1972), Parsons et al. (1972), and Chadwick (1971, 1972), although details of methods by which determinations were made was not provided. Lee and Shive (1983) and Sundell et al. (1984) published magnetostratigraphic determinations for volcaniclastic sedimentary rocks of AVS east of the main field; however, only summary polarity data were presented. Density and magnetic susceptibility data for a limited selection of AVS rocks were reported by Oliver and Christiansen (1998). Only paleomagnetic studies that yielded directional data are described below.

Shive and Pruss (1977) conducted the first detailed paleomagnetic study of the AVS. Their primary goals were to use magnetostratigraphy as a correlation tool for AVS rocks and to provide an Eocene reference pole for the North American apparent polar wander (APW) path. They sampled 39 sites from 34 basaltic lava flows at six localities along the eastern boundary of YNP (Fig. 1B). The flows were mostly from the Wapiti Formation and overlapping Trout Peak Trachyandesite, which are part of the c. 49–48 Ma Sunlight Group. Samples were mostly collected as oriented block samples (3–5 samples/flow), but three flows were sampled using a portable drill (5 samples/flow). All samples were subjected to alternating field (AF) demagnetization in fields up to 100 mT, with site-mean directions determined from the AF step that minimized within-site dispersion (most site means calculated at the 10 or 20 mT level). No samples were thermally demagnetized, but thermomagnetic determinations indicate that the carrier of magnetization is typically low-Ti titanomagnetite. Site-mean directions, mostly calculated from the 10 or 20 mT demagnetization level, were well-defined to poorly-grouped with α_{95} values that ranged from 6° to 170°, and were of normal or reversed polarity. Most sites sampled were obtained from highly exposed ridgelines and

Fig. 1. A) Location of the Absaroka Volcanic Field and other early Tertiary volcanic (green) and intrusive centers (red) in the northwestern United States. Generalized ages for selected volcanic and intrusive centers from Luedke (1994), Duke (2009), and Harlan (2006); B) generalized geology of Tertiary volcanic and intrusive rocks of the Absaroka Volcanic Supergroup (modified from Feeley and Cosca, 2003). Circles show locations of paleomagnetic sites sampled in this study. Also shown are approximate locations of paleomagnetic studies of Shive and Pruss (1977), Nyblade et al. (1986), and French and Van der Voo (1978b).

the large dispersion in site-mean directions (51% with $\alpha_{95} > 15°$) was attributed to incomplete removal of the effects of lightning strikes during demagnetization, as well as the small number of samples used in calculating the site-mean directions. Excluding sites with α_{95} greater than 30°, sites that were interpreted to have anomalous directions, or had less than 3 samples, they calculated a group-mean direction of declination (D) = 349.9°, inclination (I) = 61.8° (k = 14.7, α_{95} = 8.4°; 13 normal, 6 reversed sites) (Fig. 3), with a pole at 83.5°N, 177.4°E (K = 10.2, A_{95} = 10.1°).

In order to contribute to a better refinement of the Tertiary North American apparent polar wander path, French and Van der Voo (1978a,b) presented preliminary results from Tertiary basalts, dacites, and welded tuffs from northern YNP, most of which are from the Washburn Group and two sites from the Sunlight Group. They reported results from 122 samples from 11 sites (Fig. 1B) as virtual geomagnetic pole positions (VGPs). Examination of their data and recalculation of site-mean directions from the VGPs indicates that four sites yielded either very large α_{95} values (>20°) or anomalous directions. One site reportedly yielded a reverse polarity site mean (French and Van der Voo, 1978a), but which site was not specified. Plotting of acceptable site-mean directions in direction space yields an in situ group mean of D = 355.0°, I = 48.7° (k = 10.2, α_{95} = 19.9°), but with a high degree of scatter and a streaked distribution (Fig. 3). Although these results have never been formally published, they are cited by some workers (Moose and Acton, 1993; Hiza, 1999). Complications in the data set and uncertainty as to what units were sampled make assessing the reliability of these results problematic.

Nyblade et al. (1986) studied three ~70-m-thick basalt flows within the upper part of the Wapiti Formation (Sunlight Group) 20 km southeast of YNP (Fig. 1B) in order to use paleomagnetism and one-dimensional conductive cooling models to assess the magnitude of Eocene paleosecular variation. They collected 189 samples from 62 separate horizons within the flows, with each sample subjected to detailed progressive thermal or alternating field demagnetization. Site-mean directions were well-defined (α_{95} values of 2° to 4°) and all flows were of reversed polarity. The top and bottom flows yield identical site-mean directions, but the middle flow gave a significantly different direction (angular separation ~30°). Combining the three sites yields a mean of D = 168.8°, I = −51.2° (k = 27.3, α_{95} = 24.0°) (Fig. 3).

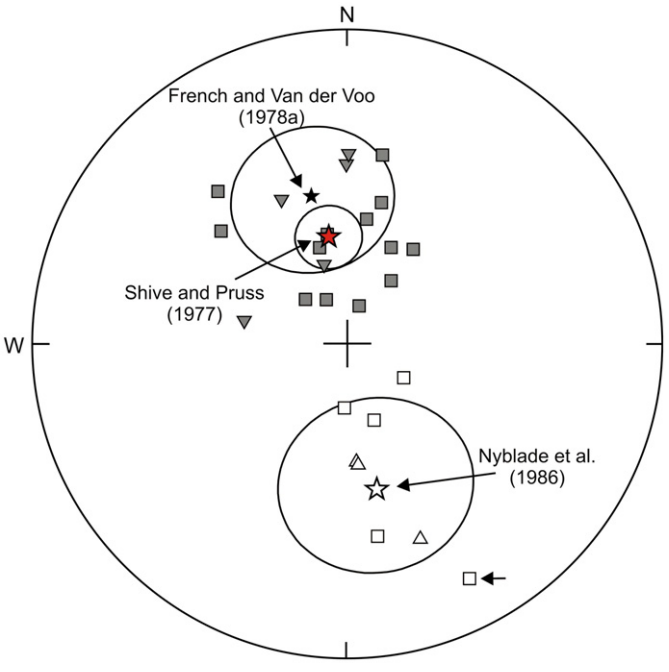


Fig. 3. Equal-area projections showing paleomagnetic site- and group-mean directions (stars) from previous paleomagnetic studies of the Absaroka volcanic rocks by Shive and Pruss (1977) (squares), Nyblade et al. (1986) (upward-pointing triangles), French and Van der Voo (1978b) (downward-pointing triangles). The data of Shive and Pruss (1977) have been filtered to exclude sites with $\alpha_{95} > 15°$ and/or anomalous directions. Data from French and Van der Voo (1978b) were obtained by recalculating site-mean directions from virtual geomagnetic pole positions. Filled symbols are projections on the lower hemisphere, open symbols show projections on the upper hemisphere.

Harlan et al. (1996b) reported paleomagnetic results from the 51.6 to 48.5 Ma Independence volcano of the Washburn Group, located ~20 km north of YNP (Fig. 1B). Eight sites were sampled, seven from the central granodiorite stock and one from an andesite plug that intrudes the stock. All samples were subjected to detailed AF or thermal demagnetization and analysis using vector diagrams and principal components analysis (Kirschvink, 1980). Site-mean directions from eight sites were well-defined and gave an in situ group mean of D = 159.4°, I = −64.0° (k = 397, α_{95} = 2.8°), with a VGP at 75.5°N, 169.6°E (K = 189, A_{95} = 4.0°). Based on geologic relations at Independence volcano, they argued that there was no need to apply tilt corrections to their results.

4. Paleomagnetic sampling and laboratory methods

Paleomagnetic samples for this study were collected from 30 sites in Tertiary volcanic, volcaniclastic, and intrusive rocks exposed in northeastern YNP, along U.S. Highway 14 east of YNP, and north of YNP in Hyalite Canyon (Fig. 1B). In Hyalite Canyon, two sites were collected in canyon-filling, columnar-jointed basalt flows of the Hyalite Peak Volcanics of Chadwick (1982), which are correlative with the Washburn Group. Within YNP, formal stratigraphic units from which samples were collected include the Elk Creek Basalt (Washburn Group; four sites in three flows) and basalt flows of the Crescent Hill Basalt (Sunlight Group; four sites in two flows), the Trout Peak Trachyandesite (Sunlight Group; two sites in different flows), the Lost Creek Member of the Sepulcher Formation (Washburn Group; four sites), fine-grained sedimentary rocks of the Langford Formation (Thorofare Creek Group; one site), and a lava flow from the Lamar River Formation (Washburn Group; one site). Two sites were collected from volcanic conglomerates from the Langford Formation of the Sunlight Group for conglomerate tests (Smedes and Protska, 1972). Eleven sites were collected from seven dikes or irregular intrusions that cut volcaniclastic strata of the Sunlight and Thorofare Creek Groups. Also reported are summary site-mean data for Independence volcano (Washburn Group) previously reported by Harlan et al., 1996b, as well as newly obtained rock magnetic data from samples from that study. Geochronologic data, map relationships, and correlation with units of known age suggest that most rocks sampled in this study were emplaced between 48 and 52 Ma.

Individual samples were obtained using a portable drill and were oriented prior to extraction using magnetic/solar compasses and clinometer; azimuths determined by magnetic and solar compasses usually differed by less than 3–5°. Typically 7 to 10 samples were collected at each site, with samples spread out over an area of 2 to 5 m^2. In the laboratory, each sample was cut into one or more 2.5 cm by 2.2 cm specimens. The natural remanent magnetization (NRM) of the specimens was measured using spinner or superconducting rock magnetometers. Each specimen was subjected to detailed progressive alternating field (AF) or thermal demagnetization procedures. Demagnetization results were analyzed using orthogonal vector diagrams, stereographic projections, and normalized intensity decay plots. Characteristic and secondary components of magnetization, recognized by collinear demagnetization paths on vector diagrams, were calculated using principal components analysis (Kirschvink, 1980). For samples in which demagnetization failed to isolate a single characteristic direction or stable endpoint but yielded a curvilinear demagnetization trajectory due to overlapping coercivity and/or unblocking temperatures (Halls, 1976), we calculated demagnetization circles using the method of

McFadden and McElhinny (1988). Site-mean directions consisting of linear data were calculated using statistics from Fisher (1953); those based on a combination of linear and great circle data were calculated using the maximum likelihood approach method of McFadden and McElhinny (1988).

Identification of magnetic carriers was based on the response of individual samples to AF and thermal demagnetization experiments and by rock magnetic experiments, including acquisition of a saturation isothermal remanent magnetization (SIRM), AF demagnetization of NRM, anhysteretic remanent magnetization (ARM) experiments, and by backfield demagnetization of the SIRM. Thermomagnetic measurements were made using low-frequency magnetic susceptibility versus temperature determinations, with estimates of Curie temperatures obtained using the second derivative method of Tauxe (1998).

5. Results

5.1. NRM and susceptibility data

NRM intensities for specimens from the AVS and Independence volcano rocks ranged from 6.3×10^{-5} A/m to 1.47×10^{2} A/m, with an average value of 4.51 A/m. Because the intensity values span several orders of magnitude, and the tendency for the arithmetic mean to be skewed towards higher values by a very small number of samples with very high intensities (>20 A/m, possibly due to contamination by isothermal remanent magnetizations caused by lightning strikes) (Fig. 4A), the geometric mean of 0.49×10^{-1} A/m is the most appropriate estimate for the mean intensity of our samples. Specimens from the Lost Creek Member of the Sepulcher Formation were significantly weaker (average intensity $=5.84\times10^{-3}$ A/m) than those from mafic and intermediate composition AVS rocks. Volume susceptibilities determined from a subset of the samples analyzed in this study ranged from 6.27×10^{-5} (paramagnetic) to 2.34×10^{-1} SI (strongly ferromagnetic; Clark, 1999), with the Lost Creek Tuff samples again yielding the lowest overall values (average susceptibility $=1.03\times10^{-4}$ SI; weakly ferromagnetic) (classification based on susceptibility data from Clark, 1999).

A broad range of Köenigsberger (Q) ratios was observed, ranging from 0.05 to 68.2 (Fig. 4B), with an average value of 4.14 and geometric mean of 1.21. The Q value, is commonly used as a measure of the relative importance of the NRM (J_R) to the induced magnetization (J_I) caused by the present-day field in studies of the total magnetization of a rock for interpreting magnetic anomalies (i.e., $Q=J_R/J_I$) and as an estimate of the relative stability of the remanent magnetization in a rock. For magnetic modeling, high Q values (i.e., $Q>10$) indicate that the total magnetization (i.e., $J=J_R+J_I$) of a rock is dominated by characteristic remanent magnetization of the sample (ChRM). For samples with low Q (i.e., $Q<0.5$), the remanence is negligible with respect to the induced magnetization and the total magnetization is thus parallel to that of the present-day geomagnetic field (Clark, 1997; McEnroe et al., 2009). Intermediate values of Q (0.5 to 10) reflect an increasing contribution of the ChRM to the total magnetization of a rock and require that ChRM directions be taken into account when modeling aeromagnetic anomalies (Clark, 1999; McEnroe et al., 2009). In general, very low Q values are common in rocks dominated by multidomain magnetite, whereas higher Qs reflect an increasing percentage of magnetic grains of pseudosingle and single domain character. The Q values from lavas sampled in this study were mostly close to or significantly greater than unity (range: 0.07 to 20.8, geometric mean = 1.79) and are typical of quickly cooled mafic to intermediate composition volcanic rocks (Clark, 1999). Q values for intrusive rocks displayed significantly greater variability (0.05 to 68.2), but a geometric mean of 0.92. The lowest Q values for the intrusions were generally obtained from rocks that were of intermediate composition and medium- to coarse-grained, such as those from Independence volcano, and some samples from the Lake Butte intrusion. The Lost Creek Tuff also displayed a wide range of variation in Q (0.06 to 18.3), with a geometric mean of 0.45. The one

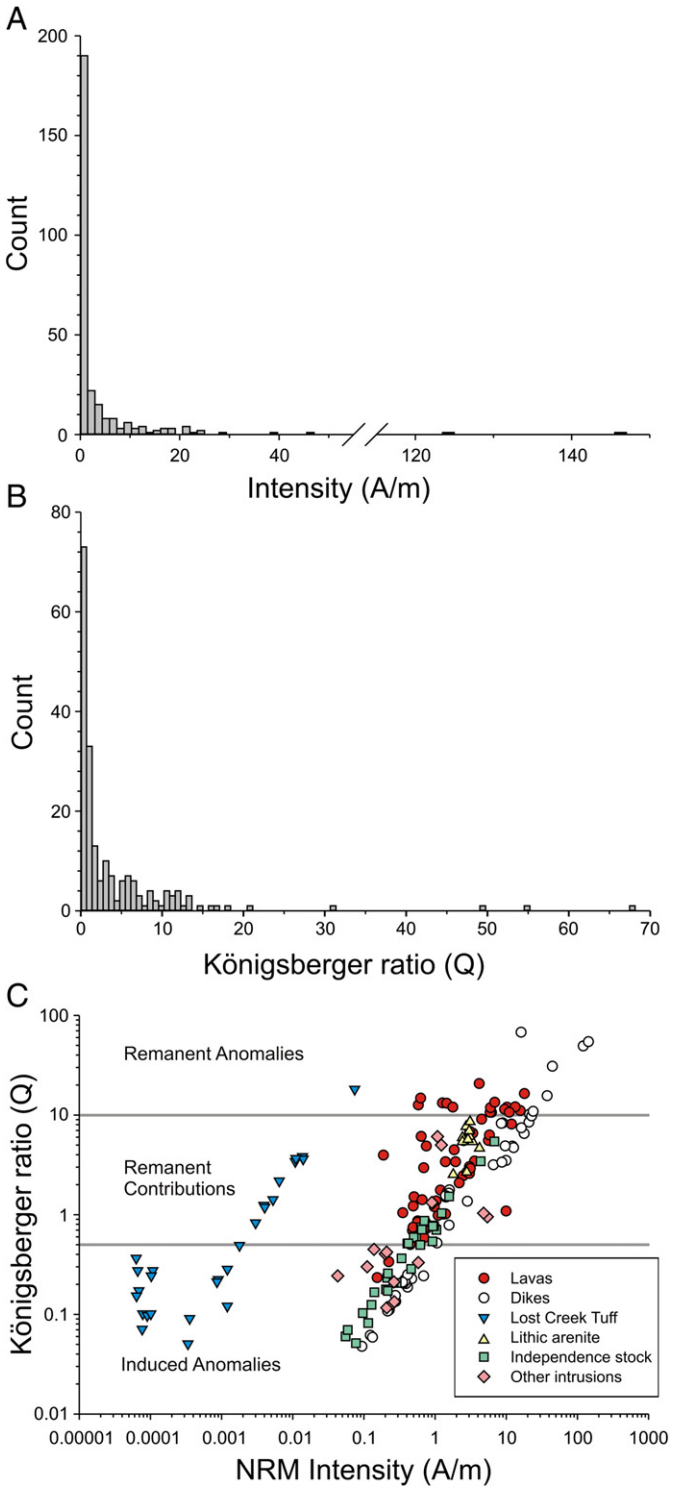


Fig. 4. Bulk magnetic properties for samples from the Absaroka Volcanic Supergroup and related intrusive rocks. A) Histogram of natural remanent magnetization (NRM) intensity (A/m) of individual samples. B) Histogram of Königsberger (Q) ratio values. C) Log–log plot of Königsberger (Q) ratio vs. NRM intensity. Filled circles, lava flows; open circles, dikes; downward-pointing triangles, Lost Creek Tuff; upward-pointing triangles, lithic arenite sedimentary rocks from the Langford Formation; squares, intrusive rocks of Independence stock; diamonds, other irregular intrusive rocks. Fields showing the relative contributions of induced and remanent magnetization components to magnetic anomalies are from McEnroe et al. (2009).

site sampled from AVS sedimentary rocks yielded Q values with a geometric mean of 5.64.

For AVS rocks there is a correlation between Q and NRM intensity (Fig. 4C), with most samples plotting in fields that indicate that the ChRM may significantly contribute to aeromagnetic anomalies observed by Finn and Morgan (2002). Most AVS volcanic and intrusive rocks are strongly magnetized and the Q values suggest that their magnetizations are dominated by pseudosingle and single domain magnetite. However, the lower Q values in the reversely magnetized intrusive rocks of the Independence volcano (geometric mean = 0.38) suggest that the origin of the pronounced magnetic high observed at this volcanic center (Peterson, 1983) may be due in part to induced magnetizations and/or a strong, positively magnetized source at depth. Similar positive anomalies are observed at other AVS intrusive centers in the YNP region, such as the Emigrant stock (Peterson, 1983) and the Lake Butte intrusion (Finn and Morgan, 2002) and may have similar origins.

5.2. Carriers of remanent magnetization

AF and thermal demagnetization behavior and rock magnetic experiments indicate that the primary carrier of magnetization in AVS rocks is low-Ti titanomagnetite. Median destructive fields during AF demagnetization for most samples are 20 to 80 mT, and laboratory unblocking temperatures are usually 550 to 590 °C. The results are consistent with the presence of a low coercivity phase such as magnetite or cation-deficient magnetite as the primary magnetic carrier in most AVS samples. In some oxidized samples, however, a significant percentage of remanence remains after AF demagnetization in fields up to 120 to 160 mT, such as sample YNP 625-2b (Fig. 5). In thermal demagnetization, these samples were characterized by unblocking temperatures >640 °C. Together, these characteristics indicate that a high-coercivity phase such as maghemite or hematite is present in these samples. Acquisition of IRM experiments show that most samples achieve saturation at applied fields of 200 to 300 mT, consistent with the dominance of the artificial magnetization by a low coercivity phase; this is true even for oxidized samples, such as sample YNP-625-2b (Fig. 5). Backfield demagnetization of the saturation IRM gives coercivity of remanence values of 22 to 38 mT, consistent with the presence of magnetite. Comparison of demagnetization of an anhysteretic remanent magnetization (ARM) and saturation IRM shows that the ARM is more resistant to demagnetization, which is consistent with the presence of magnetite in pseudosingle and single domain size ranges.

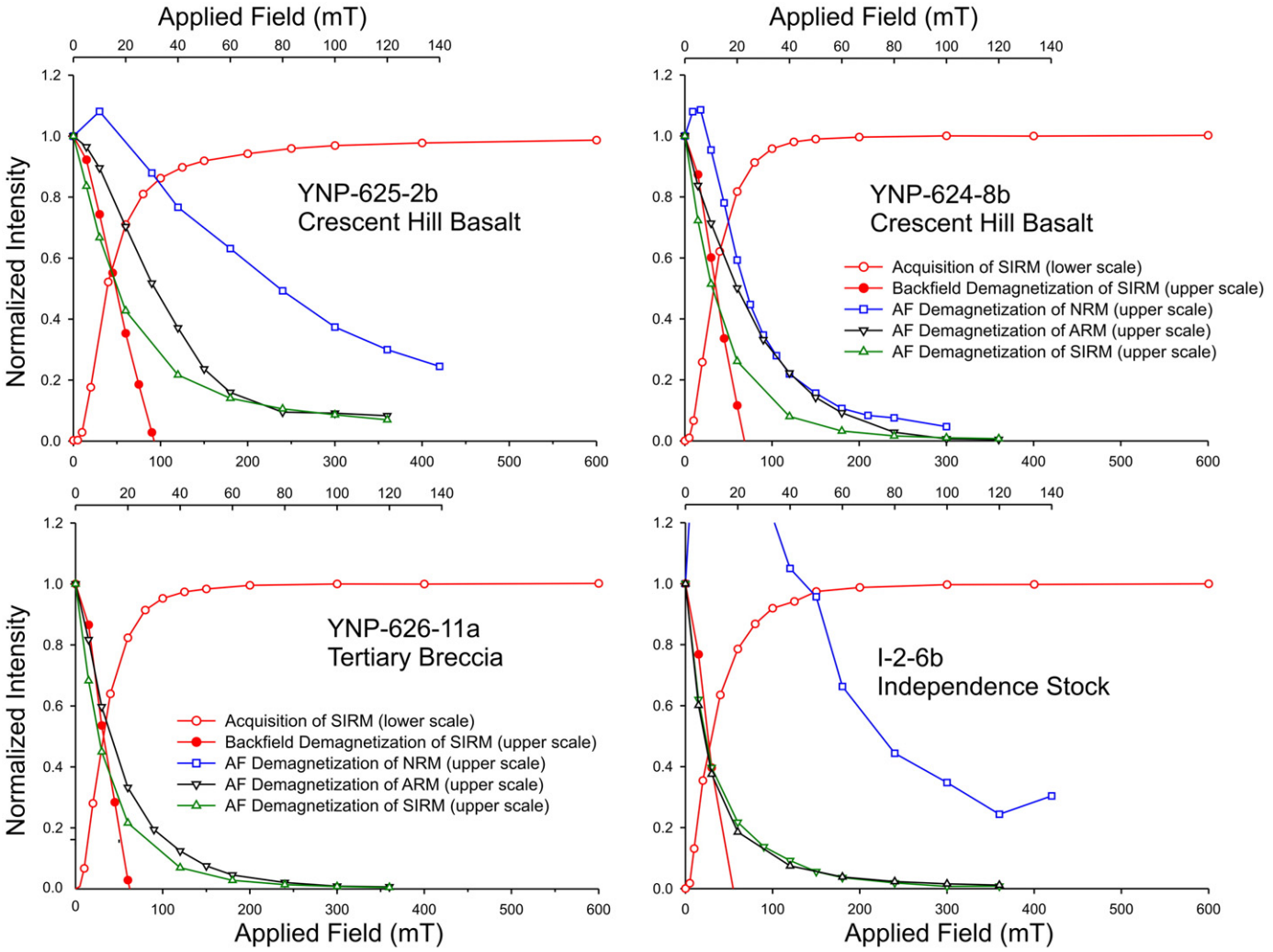


Fig. 5. Rock magnetic experiments from volcanic and intrusive rocks of the Absaroka Volcanic Supergroup and related intrusions. Open circles show acquisition of an isothermal remanent magnetization (sIRM) in peak fields up to 1200 mT (the diagram is scaled to a maximum value of 600 mT to emphasize the acquisition of the IRM at low to moderate fields) (lower scale); closed circles show backfield demagnetization of the sIRM (upper scale); square show the AF demagnetization of the sample NRM (upper scale); downward-pointing triangles show the AF demagnetization of an anhysteretic remanent magnetization (ARM) (upper scale); and the upward-pointing triangles show the AF demagnetization of the sIRM at 1200 mT (upper scale).

Low-field susceptibility versus temperature analyses (Fig. 6) show that most volcanic and intrusive samples are dominated by magnetic phases with Curie temperatures of 530 to 590 °C consistent with low-Ti titanomagnetite as the primary magnetic phase in these samples. Despite having been analyzed in a flowing argon environment, many samples show non-reversible cooling curves that are significantly less magnetic. This effect is probably due to oxidation or thermal breakdown of the samples during heating. A sample from oxidized basalt (YNP-625-5b) demonstrates the presence of two distinct magnetic phases in both heating and cooling curves: the first is poorly defined and has a Curie temperature of about 560 to 570 °C, but the overall geometry of the curve is dominated by a second phase that has a Néel temperature of about 653 °C. The relatively high susceptibility of this second phase indicates that it is probably maghemite. Hematite may also be present, judging by the red color of the sample, but is not distinguishable because the susceptibility is dominated by the other two phases.

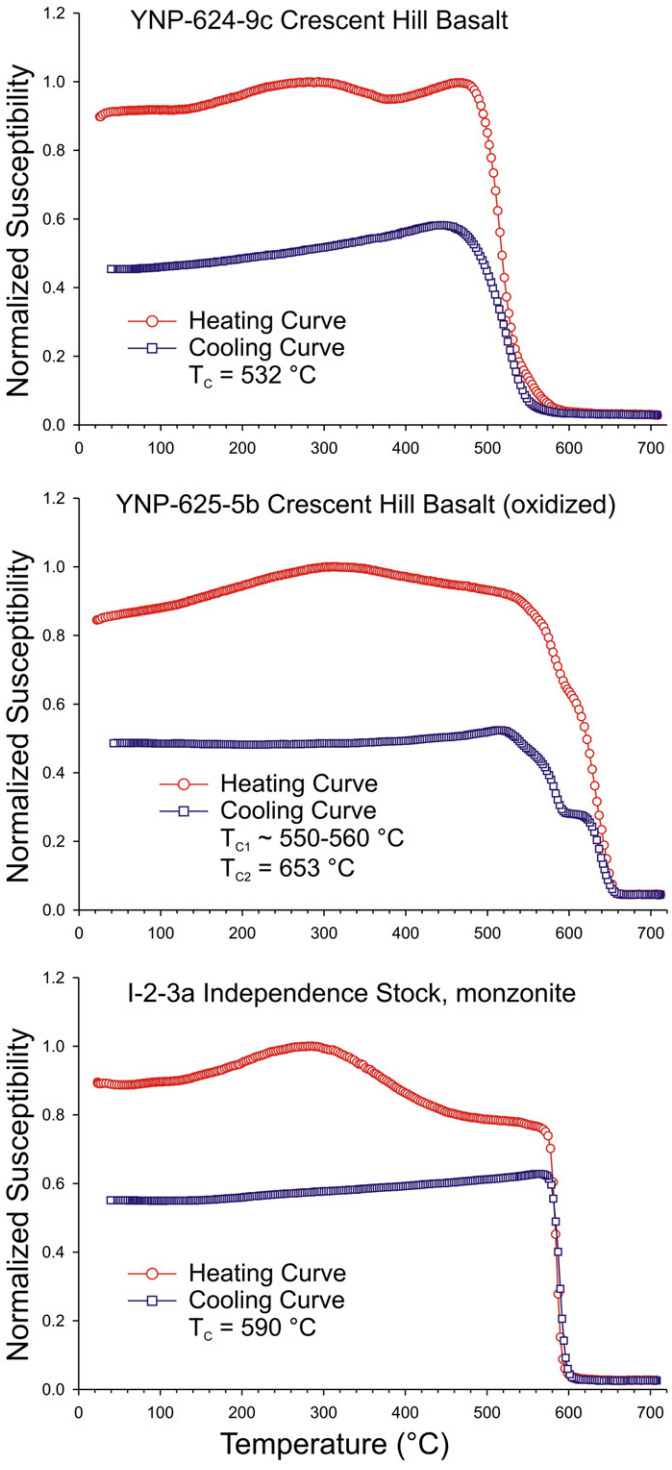


Fig. 6. Thermomagnetic (normalized low-field magnetic susceptibility vs. temperature) results from samples of intrusive and volcaniclastic rocks from the Absaroka Volcanic Supergroup. All samples were run in a flowing argon environment. T_c = Curie temperature for the heating curve determined using the second derivative method of Tauxe (1998).

Thermal demagnetization experiments indicate that low-Ti titanomagnetite is the primary carrier of stable magnetization in most samples. Normalized intensity versus temperature data illustrates highly variable behavior during demagnetization experiments (Fig. 7). Most curves show distributed unblocking temperatures with complete or nearly complete unblocking occurring at temperatures below 585 °C. In some cases, a significant fraction of the NRM remains at temperatures greater than 585 °C, indicating the presence of high-coercivity magnetic phases, such as cation-deficient magnetite or hematite. However, many of these samples become directionally unstable at higher temperatures greater than 585 °C, probably due to breakdown or oxidation of magnetic phases during the thermal experiment similar to what is observed in the susceptibility versus temperature experiments described above. Oxidized samples such as YNP625-3a show well-defined unblocking temperatures up to 650 °C and consistent directional data indicating that a significant component of the remanence is carried by hematite.

5.3. Directional data

Paleomagnetic results from most samples are straightforward, with samples typically yielding well-defined directional data following AF or thermal demagnetization as judged by the monotonic decay of the magnetization vector toward the origin of orthogonal vector diagrams (Fig. 7). Secondary directions, where present, were of north declination and moderate positive inclination or were randomly oriented. The former is probably due to a viscous remanent magnetization acquired in the present-day field, whereas the latter may be due to lightning strikes, minor alteration, or laboratory-induced secondary magnetizations. Most secondary components were removed at AF fields less than 20 mT or by laboratory unblocking temperatures less than 300 °C. The determination of characteristic directions, identified at higher AF fields or temperatures, were based on the calculation of a best-fit line from three or more collinear points on the orthogonal vector diagram; typically most specimens had maximum angular deviation (MAD) values less than 5°. Five sites yielded somewhat more complicated results with some samples showing curvilinear demagnetization trajectories indicating the presence of magnetizations of overlapping coercivity or unblocking temperature spectra. Remagnetization circles fit to these data were well-defined with MAD values $<8°$, and the remagnetization circles were combined with linear data in order to provide the best estimate of the site-mean direction.

One site in granodiorite (site 535) and one from a basalt flow (site HY-2) failed to yield interpretable paleomagnetic results. The samples from this site are characterized by erratic directional and intensity changes during AF demagnetization, presumably due to the presence of coarse multidomain magnetite that acquires a spurious magnetization during the AF procedure. Because of their unstable behavior, they are excluded from further consideration. A site located elsewhere in the same intrusion, however, yielded well-behaved samples during demagnetization, with a well-defined site-mean direction (site 395; Table 1). Samples from site HY-2 yielded somewhat less erratic behavior during demagnetization but failed to yield well-defined linear segments on orthogonal vector diagrams and individual sample

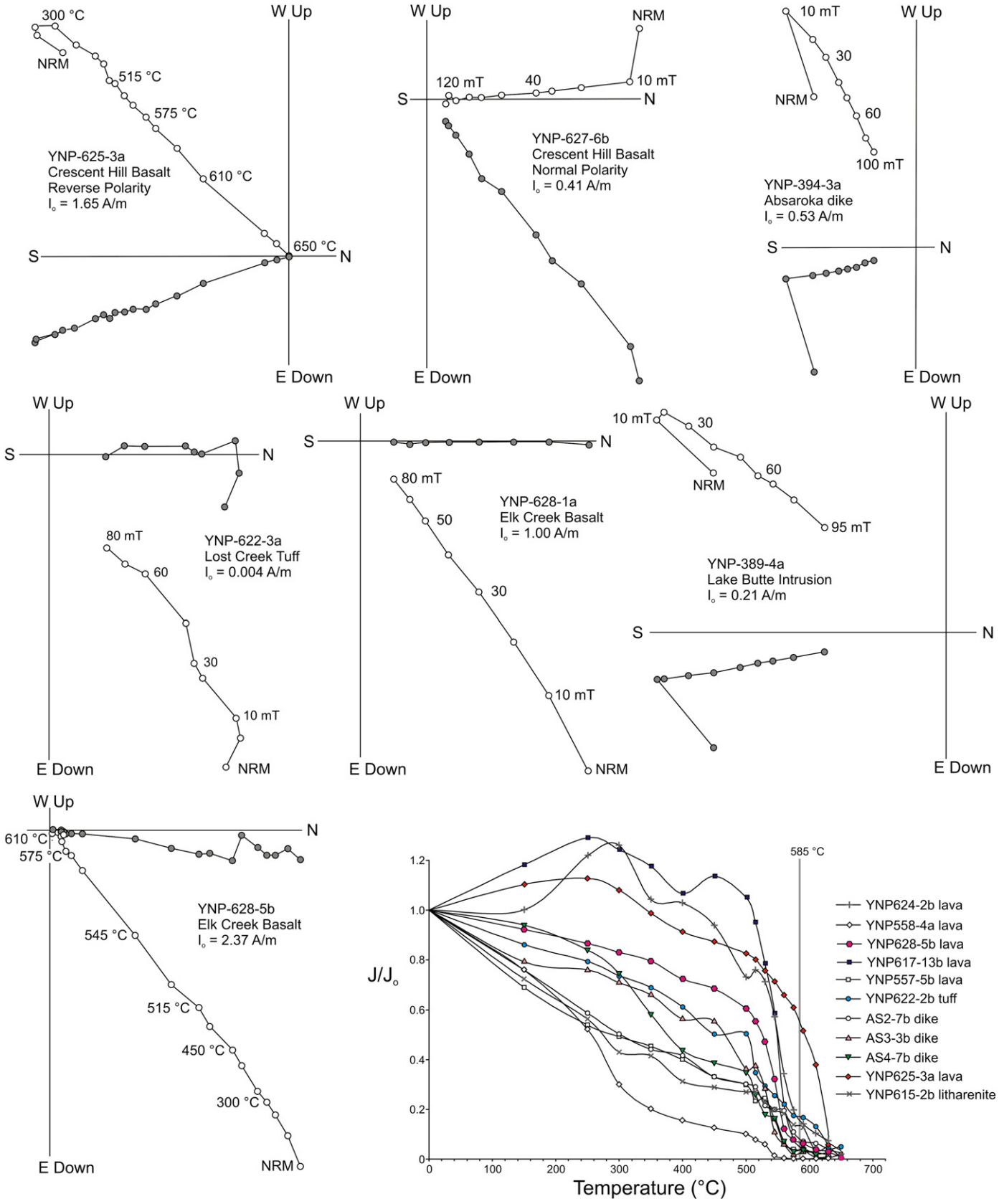


Fig. 7. In situ orthogonal vector diagrams showing examples of AF and thermal demagnetization behavior of samples from Tertiary volcanic and intrusive rocks of the Absaroka Volcanic Supergroup. Solid symbols are projections on the horizontal plane; open symbols are projections on the vertical plane. I_0 is the NRM intensity prior to demagnetization. Also shown are normalized intensity (J/J_o) versus temperature curves illustrating the laboratory unblocking of the NRM during thermal demagnetization experiments.

Table 1
Paleomagnetic site-mean results from the Tertiary Absaroka Volcanic Supergroup and related intrusions.

Map	Site no.	Geologic name	Rock type	S lat (°N)	S long (°W)	N/N_o	Dec. (°)	Inc. (°)	k	α_{95} (°)	P lat (°N)	P long (°E)
1	389	Lake Butte intrusion	Granodiorite	44.5181	110.2849	8/8	170.0	−40.8	659	2.2	67.3	94.1
2	390	Lake Butte intrusion	Granodiorite	44.5136	110.2717	7/8	171.2	−39.8	166	4.7	66.9	90.9
Lake Butte intrusion mean						*15/16*	*170.5*	*−40.3*	*293*	*2.2*	*67.1*	*92.7*
3	615	Langford Series sedimentary rocks	Lithic arenite	44.5093	110.2504	8/10	327.5	54.5	163	4.4	63.5	150.1
4	619	Langford Series[ac]	Conglomerate	44.5054	110.2535	18/19	219.2	−69.5	1.54	46.0	–	–
5	626	Langford Series[ac]	Conglomerate	44.5060	110.2540	7/10	261.9	43.4	1.44	87.1	–	–
6	395	Langford Series intrusion	Granodiorite	44.4918	110.0330	8/8	349.3	69.1	218	3.8	79.2	212.9
7	535	Langford Series intrusion[c]	Granodiorite	44.4901	110.1969	0/8	U.S.	–	–	–	–	–
8	394	Absaroka dike	Basalt	44.4952	110.2080	8/8	168.0	−53.4	140	4.7	76.0	115.2
9	396	Absaroka dike	Basalt	44.4934	110.1884	7/8	160.6	−45.6	59.3	7.9	66.6	118.0
10	AS1	Absaroka dike	Basalt	44.4698	109.6062	8/8	18.4	63.3	154	4.5	76.9	332.3
11	AS2	Absaroka dike	Basalt	44.4587	109.8445	8/8	341.0	74.5	248	3.5	70.0	222.6
12	AS3	Absaroka dike	Basalt	44.4594	109.8436	8/8	336.8	54.5	624	2.2	69.9	140.2
13	AS4	Absaroka dike	Basalt	44.4586	109.8425	10/10	356.8	75.5	539	2.1	71.7	245.5
14	618	Lost Creek Tuff Member[a]	Rhyolite	44.9456	110.4531	11/11	18.9	38.5	126	4.1	64.1	34.3
15	620	Lost Creek Tuff Member[a]	Rhyolite	44.9456	110.4531	8/10	25.3	40.2	42.9	8.6	62.1	29.5
Mean						*19/21*	*21.5*	*39.3*	*68.8*	*4.1*	*61.2*	*24.7*
16	622	Lost Creek Tuff Member[a]	Rhyolite	44.9450	110.4547	6/8	14.5	55.5	40.9	10.6	79.2	28.9
17	558	Trout Peak Trachyandesite	Trachyandesite	44.4210	110.2842	10(7)/12	350.9	61.5	108	4.8	83.2	151.3
18	557	Trout Peak Trachyandesite	Trachyandesite	44.4162	110.2787	11(9)/11	351.1	66.7	46.6	6.8	82.2	201.4
19	448	Elk Creek Basalt	Basalt	44.9556	110.4508	7/8	285.4	51.4	217	4.1	32.3	174.3
20	621	Elk Creek Basalt	Basalt	44.9444	110.4597	8/9	285.9	51.4	147	4.6	32.6	174.1
Mean						*15/17*	*285.7*	*51.4*	*186*	*2.8*	*32.5*	*174.2*
21	628	Elk Creek Basalt[a]	Basalt	44.8767	110.2158	8/8	359.7	52.3	203	3.9	78.0	71.0
22	635	Elk Creek Basalt	Basalt	44.8767	110.2158	6(3)/6	333.9	78.8	117	6.7	62.9	228.9
23	617	Crescent Hill Basalt[a]	Basalt	44.9538	110.5313	11/12	164.5	−48.6	100	4.6	61.8	126.9
24	624	Crescent Hill Basalt[a]	Basalt	44.9447	110.5219	10/11	170.2	−52.6	606	2.0	76.1	105.7
25	625	Crescent Hill Basalt[a]	Basalt	44.9450	110.5250	8/8	176.2	−47.4	444	2.4	81.2	73.3
Crescent Hill reverse mean						3/3	*170.4*	*−49.6*	*289*	*7.3*	*73.3*	*81.2*
26	627	Crescent Hill Basalt[a]	Basalt	44.9508	110.5114	9/9	343.8	30.6	115	4.8	58.4	100.2
27	616	Bunsen Peak Intrusion		44.9344	110.7069	7(4)/13	30.5	67.8	36	7.9	68.9	312.3
28	HY-1	Hyalite Peak Volcanics[ac]	Basalt	45.4722	110.9287	7/9	11.2	19.8	123	5.1	53.4	50.3
29	HY-2	Hyalite Peak Volcanics[c]	Basalt	45.4396	110.9642	0/6	–	–	–	–	–	–
30	923	Lamar River Formation	Trachyandesite	44.8210	110.44516	9(7)/9	323.5	43.3	63.3	6.6	54.7	138.2
31	I1	Independence stock[b]	Altered granodiorite	45.2219	110.2181	9/9	172.5	−59.5	481	2.4	82.6	120.7
32	I2	Independence stock	Altered granodiorite	45.2219	110.2186	9/10	161.8	−64.0	203	3.6	77.3	168.5
33	I3	Independence stock	Granodiorite	45.2217	110.2189	10/10	162.9	−60.8	56	5.9	77.2	150.5
34	I4	Independence stock	Altered granodiorite	45.2183	110.2261	9/9	151.1	−64.7	145	4.3	70.0	174.1
35	I6	Independence stock	Granodiorite	45.2169	110.2200	10/10	160.1	−63.7	49	6.3	76.0	167.4
36	I7	Independence stock	Granodiorite	45.2133	110.2183	10/10	152.0	−65.3	157	3.9	70.6	176.3
37	I8	Independence stock	Granodiorite	45.2147	110.2164	10/10	178.2	−72.3	186	3.6	72.3	178.2
Independence stock mean						7/7 sites	162.3	−64.6	637	4.3	77.6	171.9
38	I5	Andesite plug	Andesite	45.2178	110.2250	10/10	157.0	−67.1	637	1.9	73.9	184.6

Notes: map is the location of individual paleomagnetic sites on Fig. 1; Site no. is the name of the paleomagnetic site; Geologic name is the stratigraphic or map designation of the site; Rock type is the lithologic designation of the site; S lat and S long are the latitude and longitude of the paleomagnetic site, respectively; N/N_o is the number of sample directions used in calculating site-mean directions/samples measured (the number in parentheses refers to the number of stable endpoints where linear and remagnetization circle data have been used to estimate the site-mean direction); Dec. and Inc. are declination and inclination of the in situ site- or cooling-unit mean direction except where otherwise noted; k is the precision parameter of Fisher (1953); α_{95} is the semiangle of the cone of 95% confidence about the mean direction; P lat And P long are latitude and longitude of the virtual geomagnetic pole calculated from the site- or cooling-unit mean direction. Negative inclination site-mean directions were inverted to positive inclination prior to pole calculation. U.S. refers to samples unstable during demagnetization experiments as discussed in text.

[a] Site-mean directions have been corrected for tilt; all others are in situ. Tilt corrections: YNP-619: N10°E/9°W; YNP-626: N12°E/16°W; YNP-620, YNP-622, YNP-618: N13°E/4°SE; YNP-617, YNP-624, YNP-625, YNP-627: N10°E/11°W; YNP-628: N22°W/4°SW; HY-1: N3°E/6°W.

[b] Paleomagnetic data for sites I1 through I8 previously reported by Harlan et al. (1996b).

[c] Not used in the calculation of the grand-mean direction.

directions were highly scattered; consequently, a site-mean direction was not calculated.

With the exception of two sites from conglomerates discussed below, Individual site-mean directions from 26 sites are well-defined, with Fisher (1953) k values of 36 to 659, α_{95} values of 2.0° to 10.6°, and are mostly of north declination and moderate positive inclination or south declination and moderate negative inclination (Table 1; Fig. 8A). Results from multiple sites from the same flow or intrusion have been averaged to provide group-mean directions resulting in 21 individual cooling units. Incorporation of paleomagnetic results from the Independence stock along the northern part of the field (Fig. 1) and an andesite plug that cuts it (Table 1) yields 23 individual cooling units (Fig. 8B) that are treated as independent recordings of the geomagnetic field. Site-mean directions for volcanic and sedimentary strata have been corrected for minor tilt where structural data are available. However, the low variability in strike directions and magnitude of tilt corrections precludes the application of a meaningful fold test as a means to assess the long-term reliability of the paleomagnetic results. Except were otherwise noted in Table 1, results from intrusive rocks are given as in situ or geographic coordinates.

5.3.1. Reversal test

Group-mean directions based on averaging the normal and reverse polarity cooling-unit mean directions (excluding site HY-1, discussed below) yields $D=348.5°$, $I=61.4°$ ($k=18.1$, $\alpha_{95}=8.9°$, $N=16$) and $D=165.6°$, $I=-53.5°$ ($k=53.8$, $\alpha_{95}=9.2°$, $N=6$), respectively (Fig. 8C). Inverting the reverse polarity mean to normal polarity and comparing the two mean directions yields an angular difference (γ) of $6.5°\pm9.4°$ (uncertainty calculated using the method of Debiche and Watson, 1995). The two means have different k values, but using

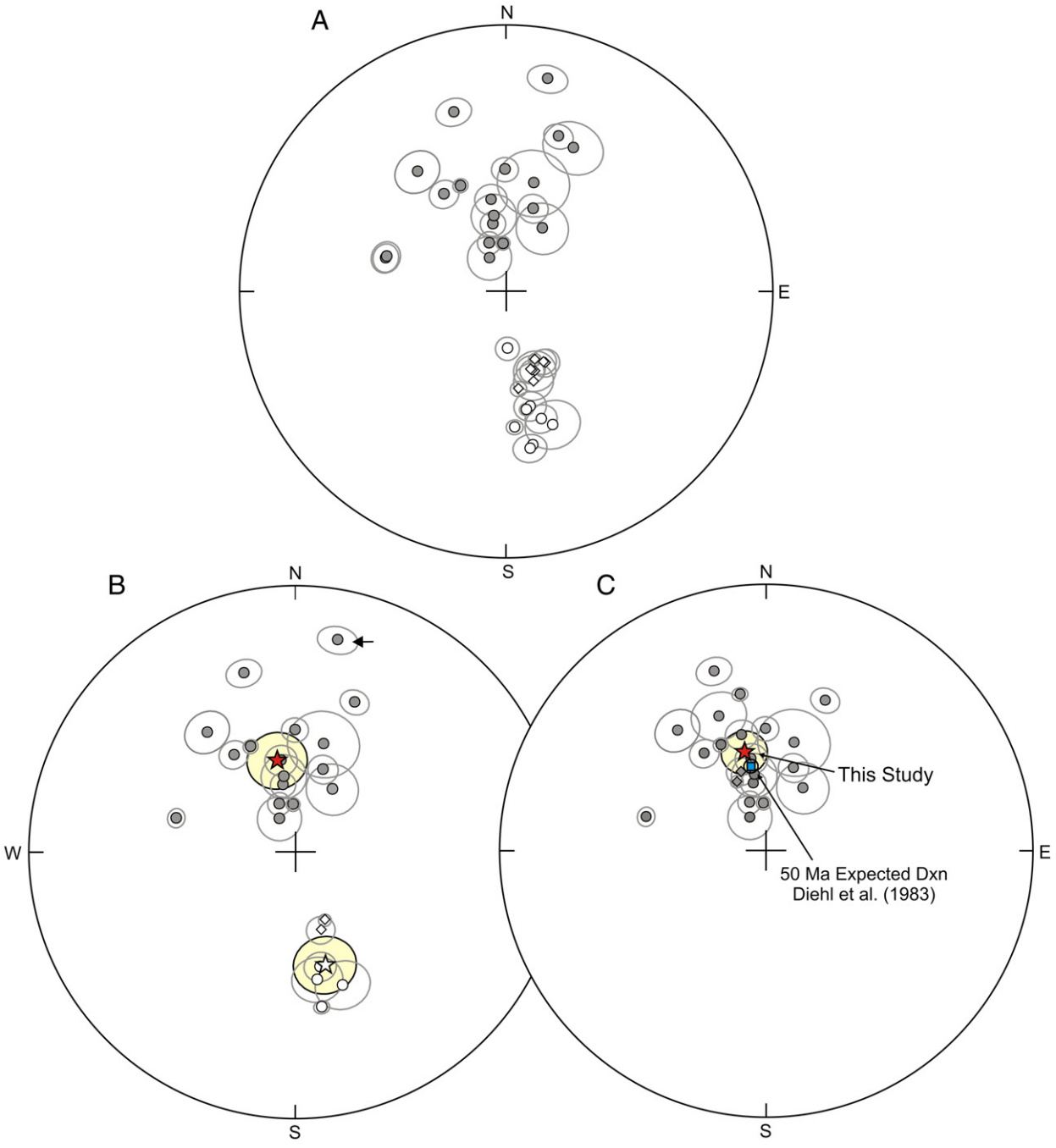


Fig. 8. Equal-area projections showing paleomagnetic results from this study. A) Individual site-mean directions from this study (circles) and Independence volcano (diamonds) and their α_{95} cones of confidence. B) Site- and cooling-unit mean directions. Also shown are normal and polarity group-mean directions and their α_{95} cones of confidence (heavy line). The site-mean direction indicated by the arrow has been excluded from calculation of the normal polarity mean direction, as discussed in the text. C) Inversion of reverse polarity site-mean directions to normal polarity and calculation of grand-mean direction (star) and α_{95} cone of confidence (heavy line). The grand-mean direction is statistically indistinguishable from the expected direction for the study area (square) calculated from the c. 50 Ma reference pole of Diehl et al. (1983). Filled symbols are projections on the lower hemisphere, open symbols show projections on the upper hemisphere.

simulation they are statistically indistinguishable as the observed γ is less than the critical value (γ_c) of 16.2° required to demonstrate a significant angular difference. Thus the reversal test is positive (class R_b) (McFadden and McElhinny, 1990).

5.3.2. Conglomerate test

Specimens from individual samples from clasts in two volcanic conglomerates from the Langford Formation yield well-defined magnetizations of a quality similar to those obtained from the volcanic and intrusive rocks, but the directions from individual clasts are highly scattered (Fig. 9). For site 619, the 18 sample directions show no apparent evidence of a preferred orientation, and have a tilt-corrected site mean of $D = 219.2°$, $I = -69.5°$ ($k = 1.54$, $\alpha_{95} = 46.0°$) (Table 1). For conglomerate tests, a nonrandom population of sample directions indicates that the observed ChRM is a secondary magnetization acquired after deposition of the conglomerate (a negative conglomerate test); a random distribution indicates that the ChRM formed before deposition (a positive conglomerate test). To test the statistical significance of this result, we compare the resultant vector of our result ($R = 6.96104$) against the critical value of R_o for a random

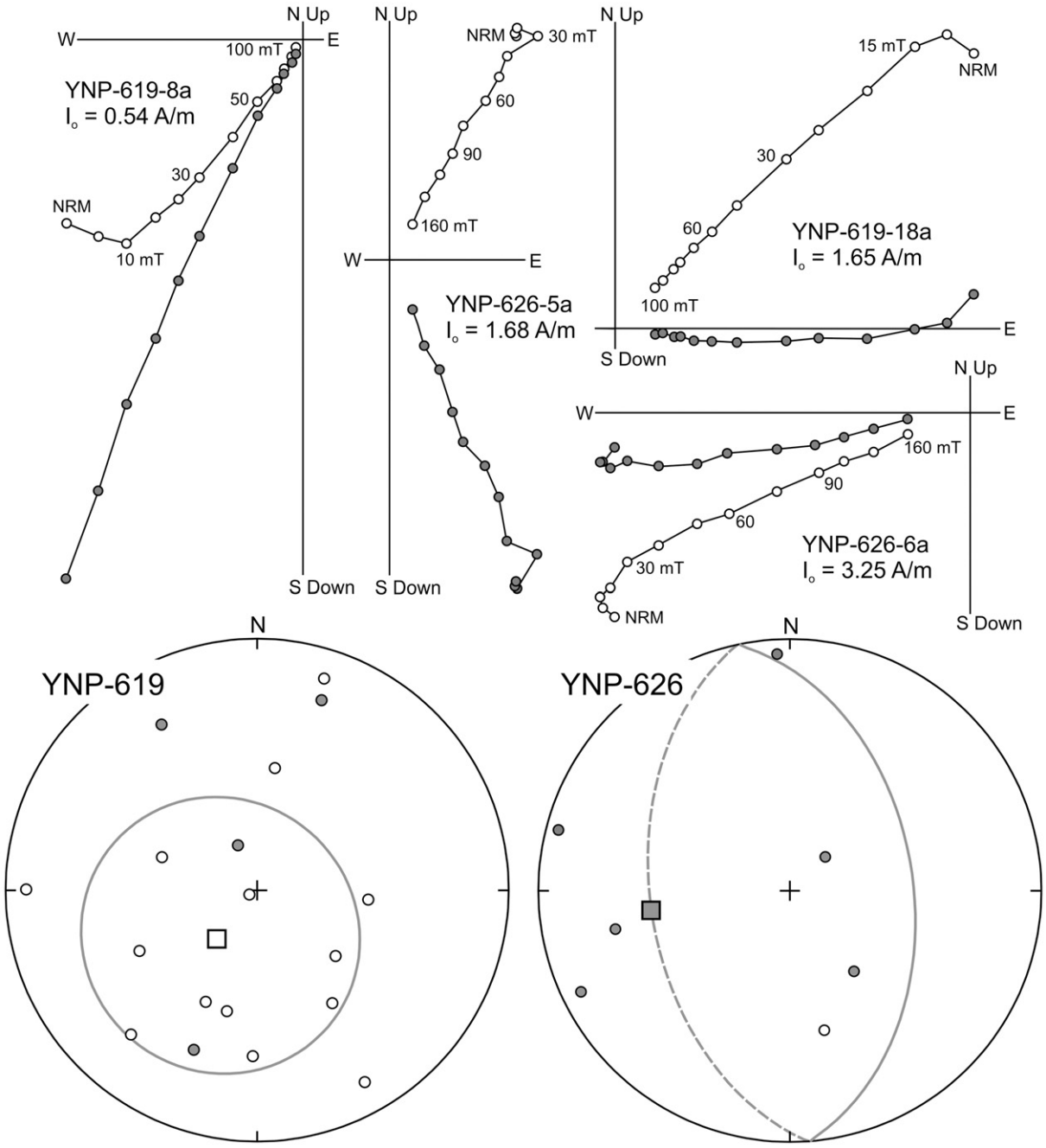


Fig. 9. Orthogonal vector diagrams and equal-area projections (stratigraphic coordinates) showing characteristic directions (circles) obtained from clasts within two volcanic conglomerates from the Langford Formation of the AVS. I_0 is the NRM intensity prior to demagnetization. Also shown are the site-mean directions (squares) and their α_{95} cones of confidence. The characteristic directions are randomly oriented thus providing a positive conglomerate test.

distribution following Watson (1956) where $R_o = \sqrt{7.185*18/3}$. If the observed R exceeds R_o, then the null hypothesis of total randomness can be rejected at the specified level of confidence (i.e., there is a nonrandom orientation of sample direction). If $R < R_o$, then randomness cannot be disproved (Tauxe, 1998). For $N = 18$, R_o is 6.84 at the 95% confidence level, which is only slightly less than our observed value of 6.96. This suggests that the distribution may be nonrandom, but given the essentially negligible difference between R and R_o, we suggest that the observed distribution of sample directions is probably random or nearly random. Similarly, the seven sample directions obtained from site 626 appear randomly distributed, with a tilt-corrected mean of $D = 261.9°$, $I = 43.4°$ ($k = 1.44$, $\alpha_{95} = 87.1°$). The observed value of R is 2.8333 is significantly smaller than critical value of 4.27 for R_o at the 95% confidence level; thus, the null hypothesis of randomness cannot be rejected resulting in a positive conglomerate test.

In contrast, lithic arenite samples from several beds of Langford Formation sedimentary rocks exposed below the conglomerate of site 619 yield a well-defined site mean of $D = 327.5°$, $I = 54.5°$ ($\alpha_{95} = 4.4°$, $n = 8/10$ samples; Table 1). The lithic arenite site-mean direction is essentially identical to those from normal polarity volcanic and intrusive rocks elsewhere in the field, and is distinct from either the time-averaged ($D = 0°$, $I = 63°$) or present-day field directions ($D = 13.2°$, $I = 69.2°$). The magnetization in these rocks is probably a post-depositional detrital remanent magnetization formed shortly after deposition of the sediments. The high dispersion in directions

from the conglomerates from both sites indicates that their clasts were emplaced randomly at temperatures well below the closure temperature for magnetic blocking, consistent with deposition from essentially cold stream or hyperconcentrated flood flow deposits. Together, the data result in positive conglomerate tests that provide evidence that the ChRM of the rocks have been stable since deposition of the conglomerate. This suggests that the remanent magnetization in the intercalated sandstone beds, as well as magnetizations observed in other nearby AVS sites are probably primary magnetizations of Tertiary age.

6. Discussion

Most samples from YNP and surrounding areas sampled in this study yield well-defined magnetizations that are well-grouped at the site level. Furthermore, stratigraphic units from which multiple sites were collected typically show good internal consistency between site-mean directions. For example, sites 448 and 621 from the Elk Creek Basalt yield identical site-mean directions (Table 1) despite the fact that they are located ~23 km apart, which suggests that they were erupted at essentially the same time and may be part of the same flow. Results from other sites in the Elk Creek Basalt yield significantly different magnetizations consistent with geologic mapping of Prostka et al. (1975a,b) that shows that 3 to 4 individual lava flows comprise this stratigraphic unit. Three sites in the Crescent Hill Basalt yield identical magnetizations that are of reversed polarity (sites 617, 624, and 625), whereas a fourth site (627) yields a magnetization of normal polarity (Table 1). The paleomagnetic data are thus consonant with field relations described by Smedes and Protska (1972) who documented the presence of two distinct partly pillowed trachybasalt flows separated by volcaniclastic and alluvial sediments of the Sepulcher Formation. The uppermost flow is of normal polarity, whereas the lowermost flow is normal. This suggests that eruption of the two basalt flows clearly spanned at least two polarity chrons, and that the Crescent Hill Basalt is not entirely of normal polarity as suggested by Smedes and Protska (1972) and Hiza (1998, 1999). In contrast, two distinct normal polarity magnetizations were obtained from the Lost Creek Tuff Member of the Sepulcher Formation. The Lost Creek Tuff Member is a sequence of rhyodacite welded ash-flow sheets. Two sites (618 and 620) sampled at two different but closely spaced stratigraphic intervals yield statistically indistinguishable site means suggesting that they were emplaced at about the same time. A third site (622) collected at another locality and significantly higher in section gives a statistically different, albeit similar direction. These differences probably record slightly different times of emplacement for individual ash-flow sheets that comprise the Lost Creek Tuff Member. The observation that the Lost Creek Tuff Member is of normal polarity contradicts the polarity assignment of Hiza (1998, 1999) who indicated that the unit was of reverse polarity.

In order to calculate a grand-mean direction we inverted reverse polarity site- and cooling-unit mean directions to a common normal polarity. Combining site- and cooling-unit mean directions that likely represent independent readings of the geomagnetic field, we obtain 22 individual directions. Site HY-1 yields an unusually shallow tilt-corrected site-mean direction ($D=11.2°$, $I=19.8°$, $\alpha_{95}=5.1°$), although individual sample directions are well-grouped at the site level (Table 1). Because the site is located in an area where the volcanic and volcaniclastic rocks are only slightly tilted (i.e., $<5°$), we assume that the site-mean direction is either a statistical outlier or represents a transitional field direction; consequently it is excluded from further calculations. A grand-mean based on the 22 acceptable site- or cooling-unit directions gives $D=347.6°$, $I=59.2°$ ($k=21.8$, $\alpha_{95}=6.8°$) (Fig. 8C). This direction is distinct from the present-day field and time-averaged directions for the study area, but is statistically indistinguishable from expected directions derived from Eocene (c. 55 to 45 Ma) reference poles for North America (e.g., $D=348.8°$, Inc.$=63.7°$, estimated $\alpha_{95}=2.5°$; Diehl et al., 1983) (Fig. 8C). Based on the behavior of the samples during demagnetization, the lack of significant or pervasive secondary magnetizations, the presence of normal and reverse polarity site-mean directions, positive conglomerate tests, and consistency of results with those from elsewhere in the region, we suggest that the observed magnetizations in our samples are primary in origin and that this collection of samples represents a faithful record of the c. 50–48 Ma geomagnetic field.

Converting site- and cooling-unit mean directions to VGPs and averaging them yields a paleomagnetic pole at 145.9°E, 80.8°N ($K=14.0$, $A_{95}=8.6°$). The angular standard deviation (S) of the pole, given by $S=81/\sqrt{k}$, is $21.6°+6.2°/-3.8°$ (upper and lower confidence limits computed following Cox, 1969, 1977). Calculating the expected VGP scatter (S_λ) for 41.8°N latitude (the calculated paleolatitude for the study area) using Model G of McFadden et al. (1991) based on paleosecular variation (PSV) studies of 5–0 Ma lava flows, we obtain a value of $16.0°\pm1.2°$. This value is slightly less than our observed value, however the scatter in our data set is strongly influenced by inclusion of a VGP (mean of sites 448 and 661) whose latitude (32.5°) is significantly less than that of the others (Table 1) and may be an outlier. Excluding this VGP yields a mean of 137.0°E, 82.6°N ($K=17.6$, $A_{95}=7.8°$, $N=21$) (Fig. 10A), which we consider to be the best estimate of a mean pole for the AVS rocks of our study. The dispersion of this pole ($S=19.3°+4.9°/-3.2°$) is slightly greater than the expected value, but the confidence intervals overlap, which suggests that enough time was sampled by our data to have adequately averaged paleosecular variation.

Combining VGPs from this study with data previously published by Shive and Pruss (1977) and Nyblade et al. (1986) allows a more refined estimate of a paleomagnetic pole for volcanic and intrusive rocks of the AVS. In reviewing the results of the earlier studies, we have filtered the data such that sites that appear to have recorded transitional field directions or whose site-mean directions are poorly determined ($\alpha_{95}>15°$) were eliminated, resulting in 21 individual VGPs. Combining their VGPs with those from our study yields a Fisherian distribution with a mean pole at 146.3°E, 83.1°N ($K=13.5$, $A_{95}=6.2°$, $N=42$) (Fig. 10B). This result, although somewhat better defined in terms of A_{95}, is indistinguishable from the pole described above (i.e., angular difference $=1.3°\pm7.5°$), but does have an angular dispersion of VGPs ($22.0°+4.0°/-2.9°$) that is greater than the expected value noted above. Incorporation of the Shive and Pruss (1977) and Nyblade et al. (1986) data thus results in somewhat more scatter than should be expected for the 5–0 Ma PSV data derived from lava flows. The reason for this excess scatter is unknown, but could reflect unrecognized problems in the data sets, such as failure to fully isolate characteristic site-mean directions based on limited demagnetization experiments performed on some of these rocks, unrecognized post-emplacement tilting of the volcanic rocks, etc. Alternatively, the values from model G based on 0–5 Ma lava flows may not be strictly applicable to judging the magnitude of secular variation for these middle Tertiary volcanic and intrusive rocks.

The YNP pole from our study, as well as that derived by combining the YNP VGPs with those of Shive and Pruss (1977) and Nyblade et al. (1986), show excellent agreement with time equivalent reference poles of Diehl et al. (1983) and Irving and Irving (1982) from cratonic North America, as well as with the synthetic 50 Ma reference pole of Besse and Courtillot (2002) (Fig. 10C), with statistically indistinguishable angular differences of $2.7°\pm5.4°$, $4.7°\pm5.6°$, and $4.3°\pm5.7°$, respectively. Our result is compatible with the argument that YNP data have fully averaged PSV and are thus a reliable record of the c. 50 Ma geomagnetic field that are of value in further refining the Tertiary apparent polar wander path for North America. Our result also demonstrates that the rocks of the AVS used in this study have not been significantly affected by post-emplacement deformation. The agreement of the AVS poles with c. 50 Ma reference poles for North America is inconsistent with the hypothesis of Moose and Acton

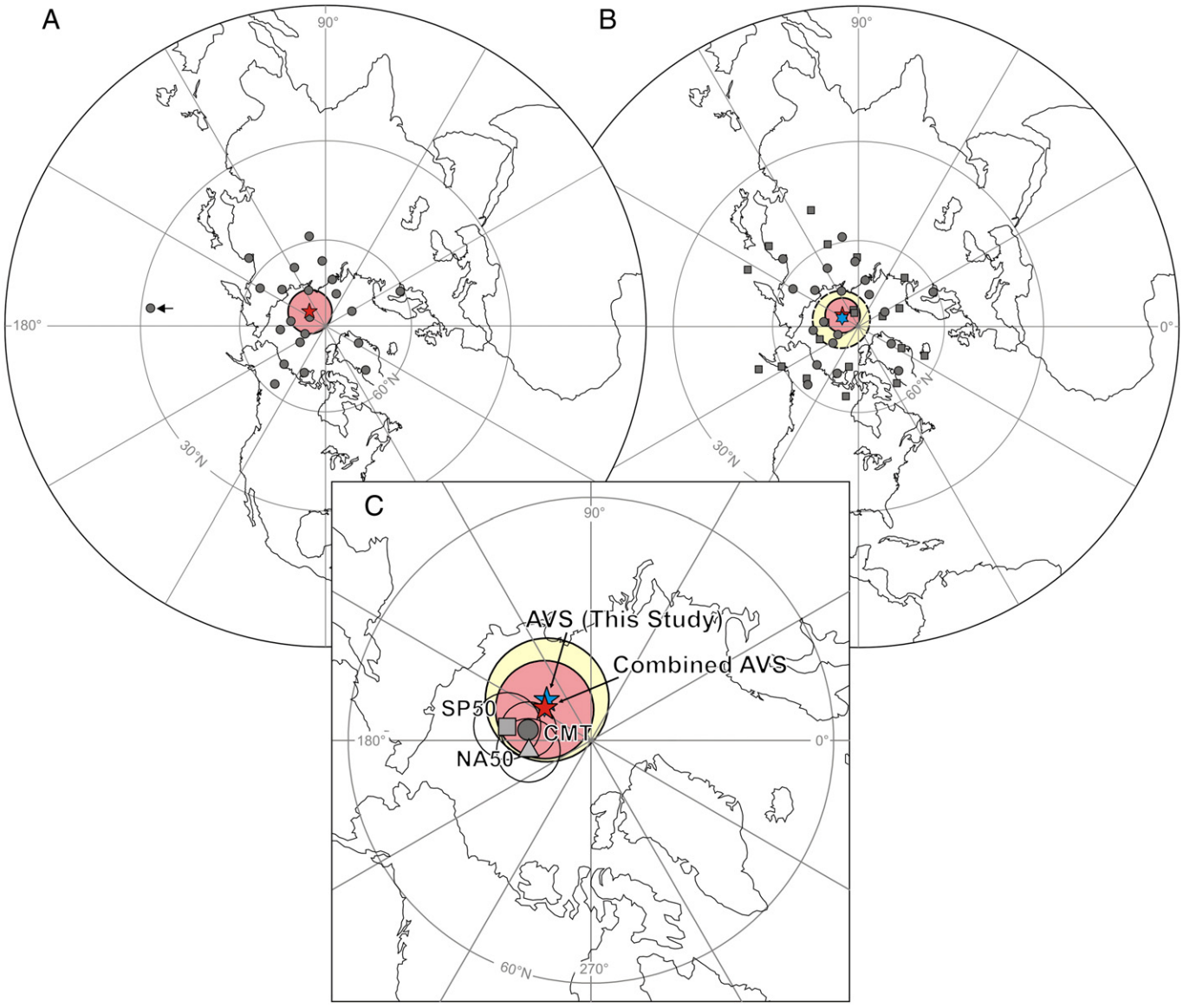


Fig. 10. Equal-area projections showing virtual geomagnetic poles (VGPs) from rocks of the Absaroka Volcanic Supergroup and related intrusions. A) VGPs (circles) from this study and mean pole (star) and its A_{95} cone of confidence. The pole calculation excludes the VGP designated by the arrow because of its low latitude relative to the other VGPs. B) VGPs from Shive and Pruss (1977) and Nyblade et al. (1986) (squares) and the mean pole (six-pointed star) and A_{95} cone of confidence (dashed line) obtained by averaging both data sets. Circles show VGPs from this study and the five-pointed star shows the mean AVS pole and A_{95} (solid line) calculated using the entire suite of VGPs. C) Comparison of the AVS paleomagnetic poles with c. 50 Ma reference poles for North America. The AVS pole is derived from VGPs listed in Table 1, as described in the text; the combined pole includes VGPs from this study, Shive and Pruss (1977), and Nyblade et al. (1986). CMT is the Tertiary reference pole from Central Montana intrusions of Diehl et al. (1983); NA50 is the 50 Ma North American reference pole of Irving and Irving (1982); SP50 is a synthetic 50 Ma reference pole determined by Besse and Courtillot (2002).

(1993) who argued that most paleomagnetic poles from the U.S. show systematic displacement consistent with clockwise rotation of the western U.S. during early Tertiary Laramide deformation caused by interaction between the North American and Pacific plates. According to Moose and Acton (1993), the apparent ($8° \pm 2°$) rotation is most readily discernable based on comparison of time equivalent poles from other plates after reconstruction to a North American reference frame. However, the agreement of the AVS pole with the synthetic 50 Ma pole of Besse and Courtillot (2002) does not support the hypothesis of significant clockwise rotation of this part of the western U.S.

7. Summary

Paleomagnetic results from volcanic and intrusive rocks of the AVS yield well-defined magnetizations that are well-grouped at the site level. Both normal and reverse polarity site means are present indicating that the AVS rocks sampled in this study have recorded multiple polarity chrons, consistent with the range of ages that represent the peak of volcanic and intrusive activity within the Absaroka volcanic field. A paleomagnetic pole calculated from data obtained during this study is consistent with results obtained by previous studies of AVS rocks and with c. 50 Ma poles from cratonic North America. A combined pole based on our results and the results of earlier studies of the AVS demonstrate that there has been little or no deformation of the region since emplacement of the volcanic and intrusive rocks. The VGPs from this study are suitable for inclusion in Tertiary reference poles for North America.

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